Sales Forecasting & Demand Intelligence

Executive Business Report

Prepared for: Head of Supply Chain & Chief Financial Officer | Data: 4 years of Superstore sales history (2015–2018)

Executive Summary

Over the past four years, sales have grown steadily and reliably rise every November and December due to holiday shopping, then dip in January and February. We built and compared three different forecasting methods and found one classic statistical model called SARIMA, most accurately predicts sales, especially during the biggest swings. Using it, we expect sales of roughly $47,000 in January, $40,000 in February, and $72,000 in March 2019, though actual results could reasonably land anywhere in the ranges shown below. We also automatically scanned four years of weekly sales for unusual spikes or drops, and grouped our products into four demand types so stocking decisions can be tailored rather than one-size-fits-all. The clearest opportunity: the East region and the Furniture category are growing faster than the rest of the business and deserve additional investment, while a few products (notably Machines) are shrinking and should be de-stocked deliberately.

Key Findings from Data Exploration & Forecasting

- Technology is our highest-revenue category ($825,856 total over 4 years), but the East region shows the most consistent year-over-year growth of any region — it grew every single year, while every other region had at least one down year.
- Shipping speed is consistent company-wide: orders ship in about 4 days on average, with less than half a day of difference between the fastest region (East, 3.9 days) and the slowest (Central, 4.1 days). This is not currently a differentiator worth fixing regionally.
- Sales follow a clear yearly rhythm: a slow start each January/February, a smaller mid-year bump, and a strong run-up through November and December driven by holiday shopping.
- We tested and compared three forecasting approaches (SARIMA, Facebook Prophet, and a machine-learning model called XGBoost). SARIMA gave the most reliable predictions overall, particularly around the large holiday sales swings, so it is the model behind the forecast below.

3-Month Sales Forecast (January – March 2019)

In plain terms: we expect a normal seasonal dip after the December holidays, followed by a rebound in March. The "likely range" reflects genuine uncertainty — think of it as the band we're 95% confident the real number will fall inside, not a guarantee of the exact figure.

Month	Expected Sales	Likely Range (95% confidence)	vs. Same Month Last Year
January 2019	$46,800	$17,000 – $76,600	Slightly higher
February 2019	$40,300	$9,600 – $71,000	Comparable
March 2019	$72,200	$41,300 – $103,100	Higher

Top 3 Unusual Sales Weeks Detected

We used two independent methods to automatically flag unusual weeks — one AI-based (Isolation Forest) and one statistical (rolling Z-score). The three weeks below were flagged by both methods, making them our highest-confidence anomalies.

Week	Sales	Likely Real-World Cause
Mar 22, 2015	$37,700 (spike)	A single large bulk or corporate order — flagged as unusual by both detection methods used, our highest-confidence anomaly.
Dec 2, 2018	$36,000 (spike)	Cyber Monday / early holiday shopping surge — consistent with the strong November–December pattern seen every year in the data.
Feb 22, 2015	$200 (near-zero)	An unusually quiet week, far below the seasonal baseline — worth checking whether this reflects a genuine demand lull or a gap in order data for that week.

Product Demand Segments & Stocking Strategy

We grouped our 17 product sub-categories into four demand types based on sales volume, growth trend, volatility, and typical order size. Each group needs a different stocking approach:

Segment	Products	Recommended Stocking Strategy
Growing Demand	Copiers	Increase reorder frequency and safety stock now, ahead of rising demand. Orders are large and irregular, so keep buffer capacity for occasional big purchases.
Declining Demand	Machines	Wind down inventory deliberately: smaller, more frequent orders, consider clearance pricing on excess stock, and avoid new large purchase commitments.
High Volume, Volatile Demand	Accessories, Binders, Chairs, Phones, Storage, Tables	This is our highest-revenue group but demand swings month to month. Hold a larger safety-stock buffer and review stock levels weekly rather than on a fixed monthly cycle.
Low Volume, Stable Demand	Appliances, Bookcases, Art, Envelopes, Labels, Furnishings, Fasteners, Supplies, Paper	Demand is small and predictable. A lean, just-in-time ordering approach keeps capital free without risking stockouts.

Business Recommendations

1. Shift incremental inventory investment toward the East region and Furniture category. Both are forecast to grow well above the company average over the next 3 months (East: +150% vs. the same period last year; Furniture: +51%), the strongest upward signal of any segment we measured.

2. Build a holiday-specific stocking plan for the "High Volume, Volatile Demand" group (Accessories, Binders, Chairs, Phones, Storage, Tables). These 6 sub-categories drive the majority of revenue but swing unpredictably month to month; a fixed monthly reorder cycle risks stockouts in strong months and overstock in weak ones.

3. Begin an orderly wind-down of Machines inventory. It is the only product group with negative growth (-11% per year) in our analysis, and carries high order value — indicating a shrinking pool of large, infrequent buyers rather than steady demand. Reducing reorder volume now avoids tying up capital in a declining product line.

Risk & Limitation to Keep in Mind

This system is built on 4 years of historical data and works best at predicting normal, repeating patterns — it cannot foresee one-off shocks such as a new competitor, a supply chain disruption, or a sudden change in the economy. It should be treated as a well-informed planning aid, not a guarantee, and predictions should be revisited monthly as new sales data comes in. We'd also recommend re-checking the model after any major business change (new product lines, new regions, or a pricing shift), since it was trained entirely on our past sales patterns.